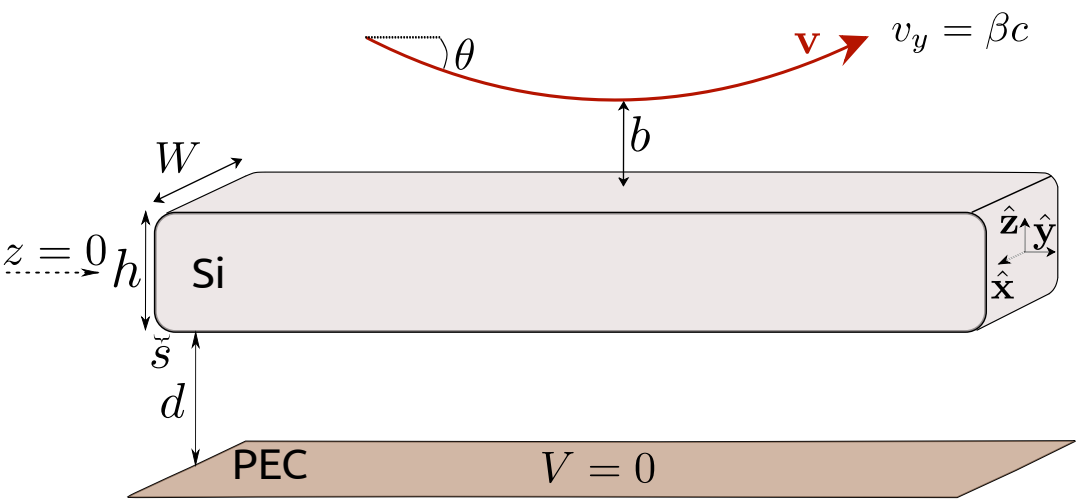


Electron-coupling-to-waveguide

$$W = 400 \text{ nm} \quad h = 300 \mu\text{m}$$

$$E_e = 200 \text{ keV}$$



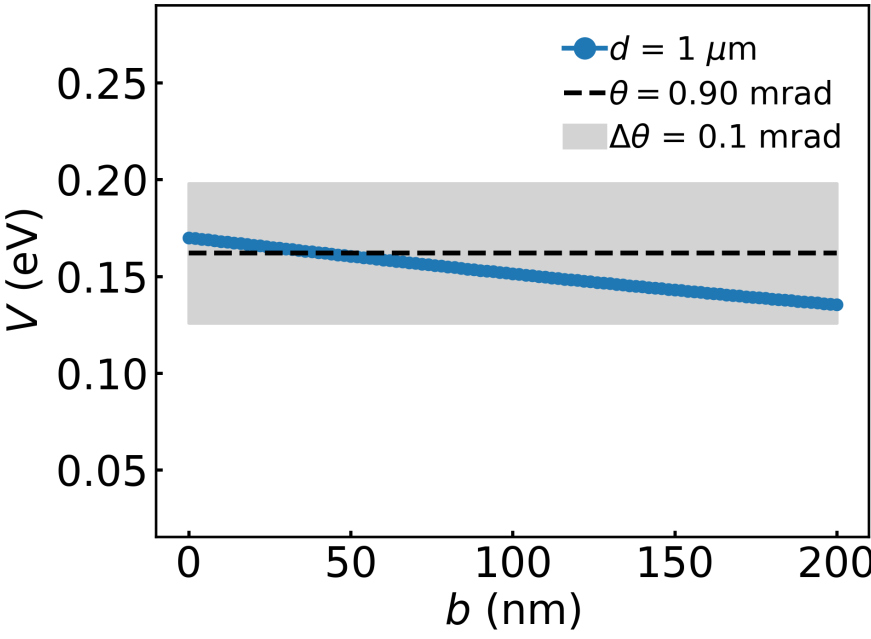
$$E_{\perp} = E_{\text{electron}} \theta^2 \text{ marked in dashed black line}$$

$$\Delta E_{\perp} = 2E_{\text{electron}} \theta \Delta\theta \text{ marked in grey}$$

For smaller d , the slop is bigger

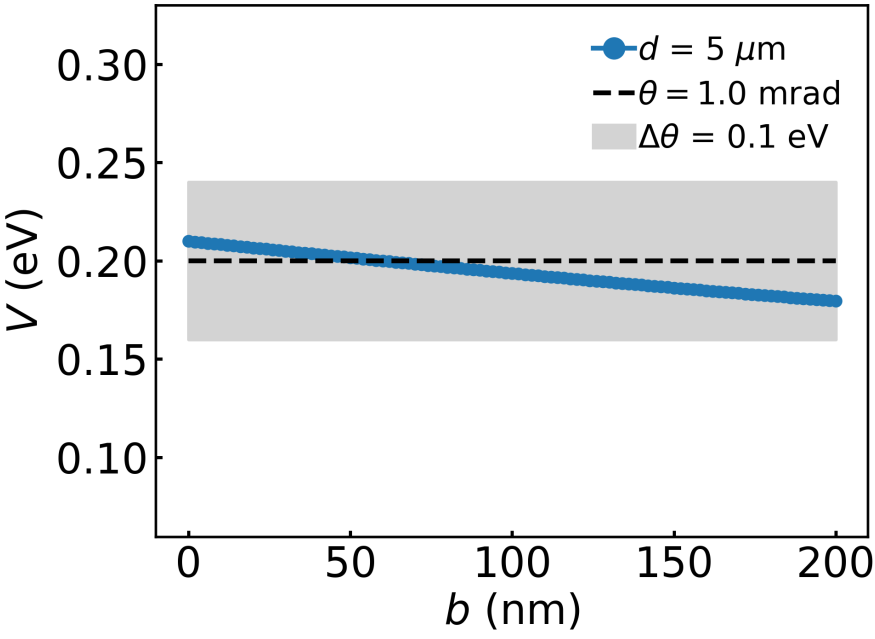
$d = 1 \mu\text{m}$ (too small)

$W = 400 \text{ nm}, h = 300 \text{ nm}, s = 20 \text{ nm}, V_0 = 0.17 \text{ eV}$



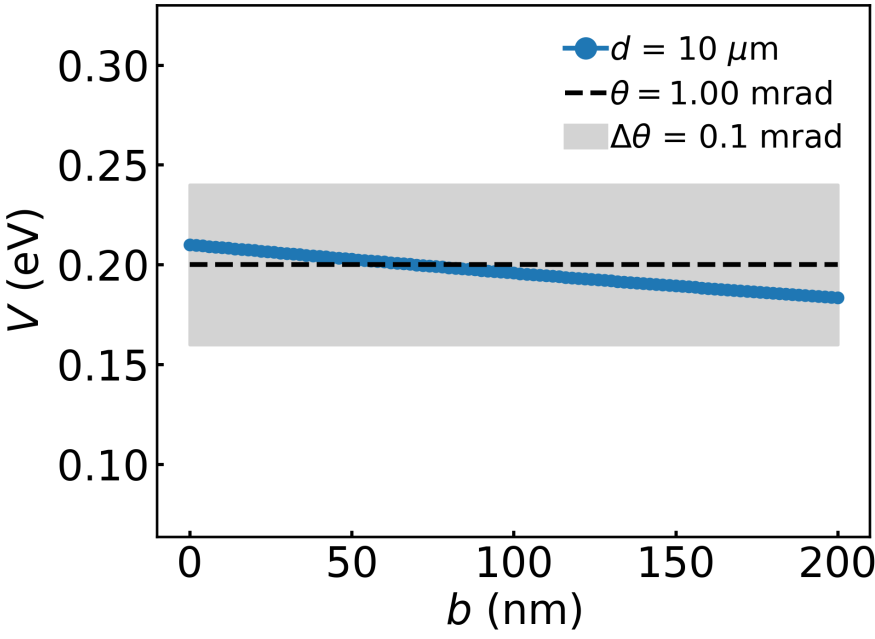
$d = 5 \mu\text{m}$

$W = 400 \text{ nm}, h = 300 \text{ nm}, s = 20 \text{ nm}, V_0 = 0.21 \text{ eV}$



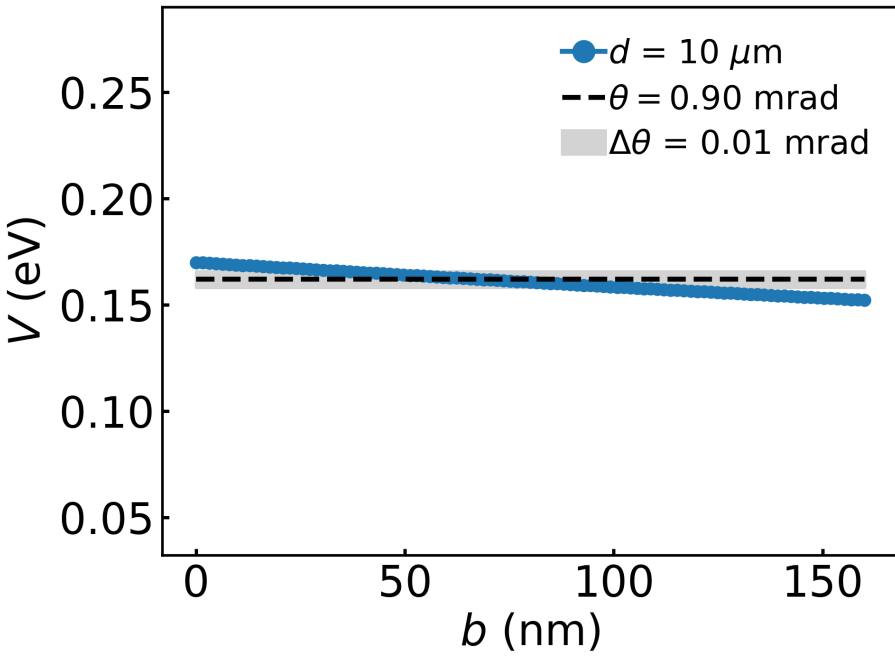
$d = 10 \mu\text{m}$

$W = 400 \text{ nm}, h = 300 \text{ nm}, s = 20 \text{ nm}, V_0 = 0.21 \text{ eV}$

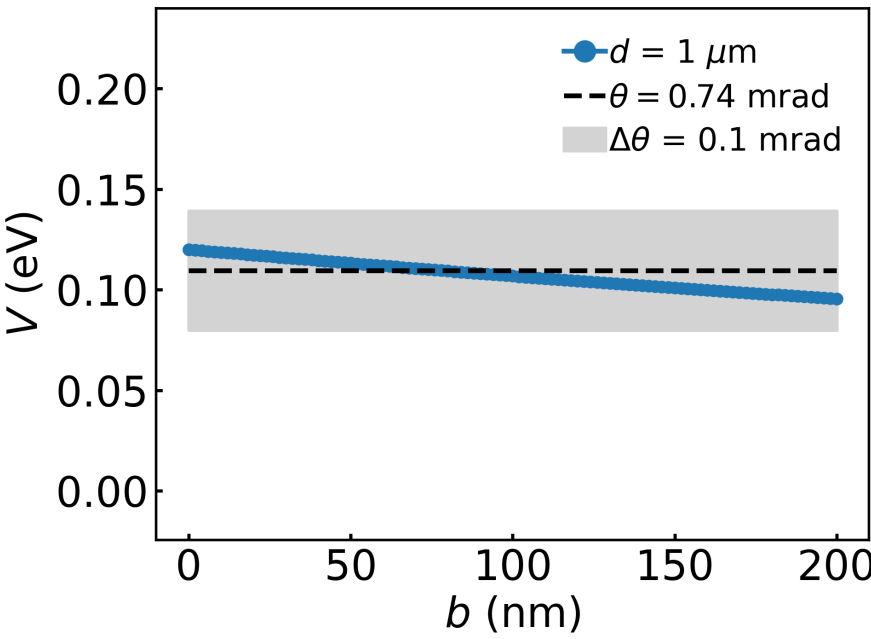


Not realistic values:

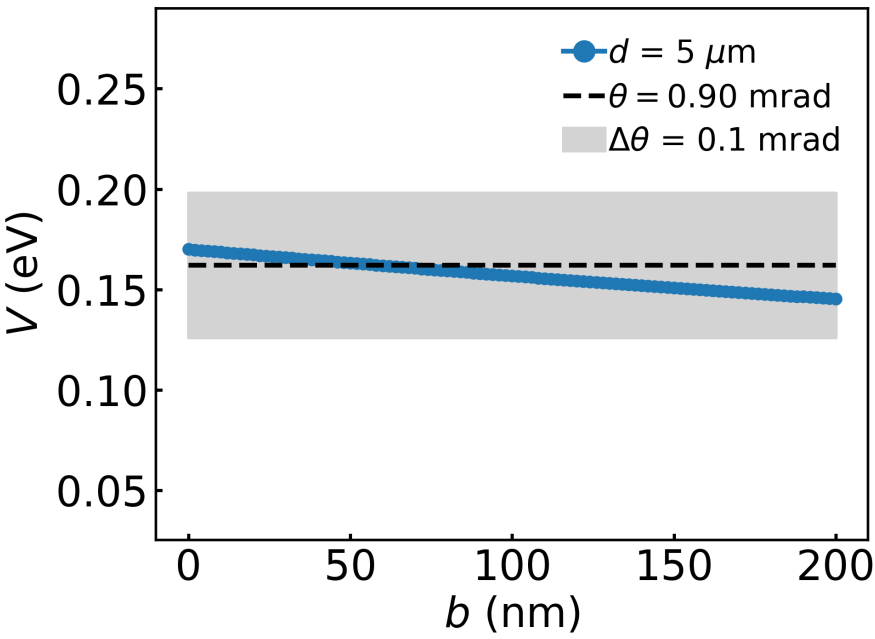
$W = 400 \text{ nm}, h = 300 \text{ nm}, s = 20 \text{ nm}, V_0 = 0.17 \text{ eV}$



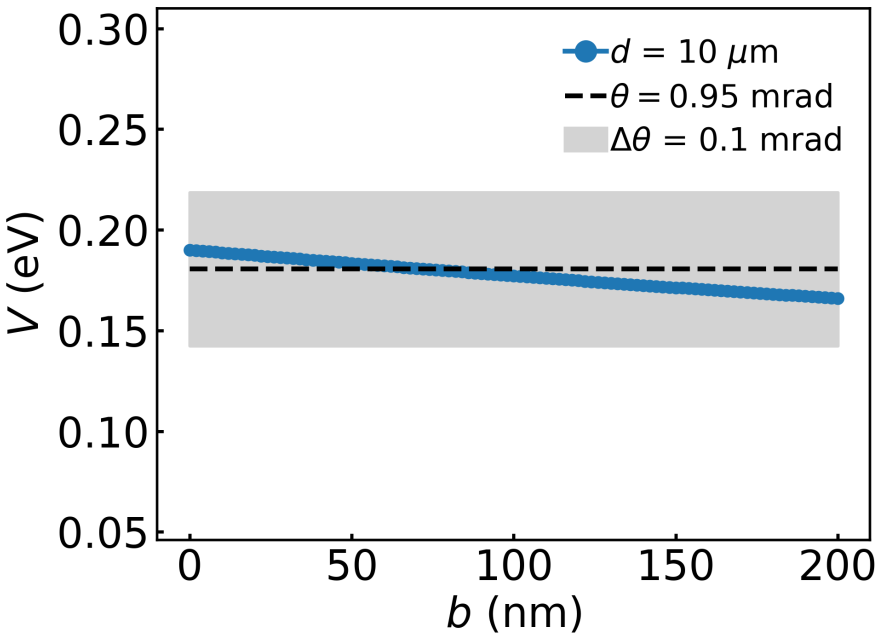
$W = 400 \text{ nm}, h = 300 \text{ nm}, s = 20 \text{ nm}, V_0 = 0.12 \text{ eV}$



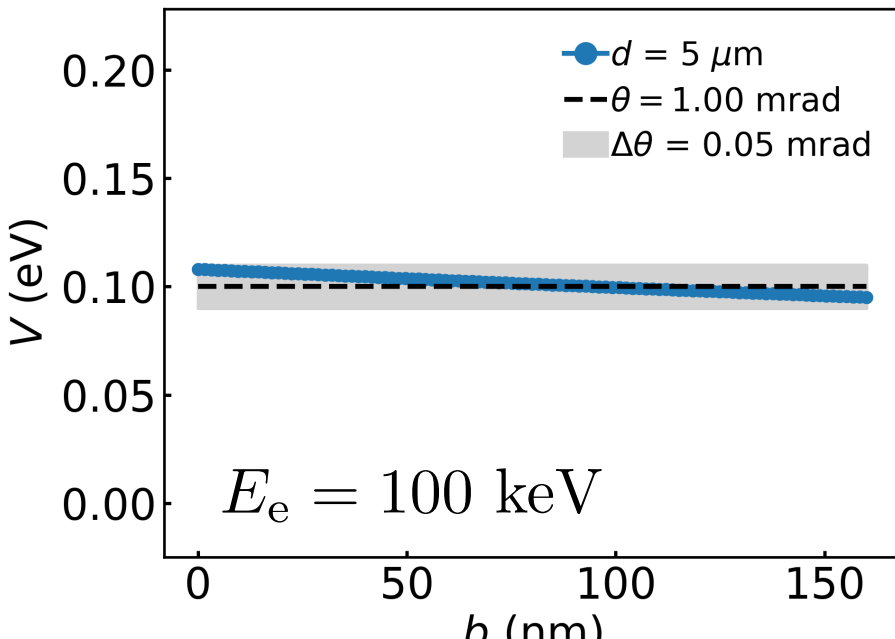
$W = 400 \text{ nm}, h = 300 \text{ nm}, s = 20 \text{ nm}, V_0 = 0.17 \text{ eV}$



$W = 400 \text{ nm}, h = 300 \text{ nm}, s = 20 \text{ nm}, V_0 = 0.19 \text{ eV}$



$W = 400 \text{ nm}, h = 300 \text{ nm}, s = 20 \text{ nm}, V_0 = 0.11 \text{ eV}$



$E_e = 100 \text{ keV}$